

Internet Architecture

- 5 Layers
 - Application (HTTP, DNS, FTP, SMTP):
 - network applications and their protocols
 - defines the way a client interacts with a given protocol
 - data in this layer is a message
 - Transport (TCP, UDP):
 - transports data from source application layer to the dest application layer
 - handles sending/receiving of raw data
 - data in this layer is a segment
 - Network (IP):
 - delivers sent transport-layer data to the receiving transport layer
 - defines sockets & the means to transport the data from transport layer
 - data in this layer is a datagram
 - Link (DSL, ethernet, â€¦):
 - moves datagrams from one node (host or router) to the next node
 - handles establishment of connections & reliable transfer between nodes
 - data in this layer is a frame
 - Physical (copper wire, fiber optics, â€¦):
 - moves the individual bits from node to the next
- IP Stack?
 - application - transport - network - link - physical
- ISO/OSI model?
 - application - presentation - session - transport - network - link - physical

Application Layer

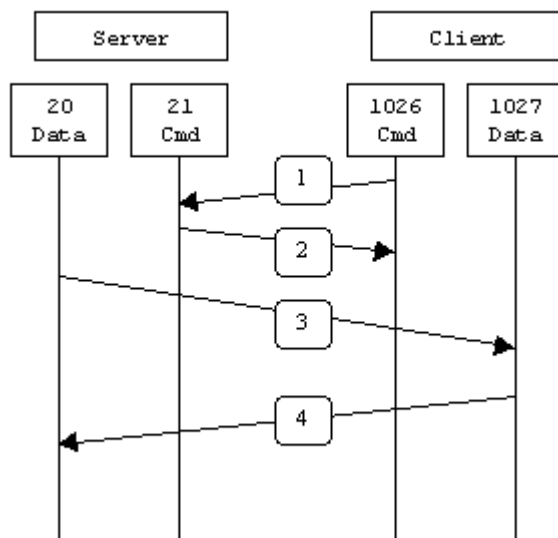
- **HTTP- Hypertext Transport Protocol**
 - Foundation of data communication for the internet
 - Protocol for webpages

- Request/response protocol
- Uses TCP for communication
- 1 port: 80
- Maintains no info about past clients
- Non-persistent HTTP connection
 - TCP connection is closed after object is sent back
 - Bad because the RTT overhead increases
- Persistent HTTP
 - Stays open for a certain amount of time while client is requesting data
- Steps:
 - client initiates TCP connection to server on port 80
 - HTTP request message is sent to server via socket
 - HTTP server receives request, retrieves it, and sends back a response
 - client gets response and closes TCP connection
- **Pull protocol**- pulls (requests) data from a server.

• **FTP- File Transfer Protocol**

- Used to transfer computer files from one host to another over a TCP-based network
- Client/Server architecture
- Separate control/data connections between client/server
- 2 ports: 20,21
- Must maintain states about the user, control connection, and local/remote directory
- Port 20
 - **Connection** port, a **non-persistent** connection that handles **authentication** of account accessing
 - used to send files
- Port 21
 - **Command** port
 - Used to take the client commands
 - **persistent** connection

- keeps track of user command, connection, current directory
- When FTP first starts:
 - client initiates TCP on port 21 and send authorization and commands to server. persist
 - Port 20 is the transfer connection. Non-persistent. The **data** is **transferred** to this port
 - Server sends one file per request, and closes the TCP after the file has been transferred (opens more as new requests come in)



• DNS- Domain Name Service

- Hierarchical distributed naming system for computer services or any resource connected to the internet or a private network
 - Root domain
 - Top Level domains
 - Second level domains
 - Computer
- Translates domain names to IP addresses
- Internet's primary directory service
- 1 port: 53

• SMTP- Simple Mail Transfer Protocol

- Protocol for email transmission
- Sender User Agent (email account) -> Sender SMTP server -> TCP Send SMTP<->Receive SMTP -> Receiver SMTP server ->Receiver User agent

- Sender -> submission -> transfer -> record -> delivery -> dest
- 2 ports: 25, 587
- **Push protocol**- pushes(sends) data from servers to other servers or clients

Transport Layer

TCP & UDP: protocols used to send data over the internet/local address to an IP address

• TCP- Transfer Control Protocol

◦ Multiplexing/Demultiplexing

- **does** require info about the data file and its contents in order to ensure connection-based transmission where delivery in order is required Reliable but slower
- Good for web-browsing/sending files
- **[source IP, source Port, dest IP, dest Port]**

- Makes sure to receive packets in order
- Most commonly used protocol on the WWW
- **Check-sum length???**
- Does binary addition and stores it
- To check if none of the data is corrupt in between packets

• UDP- User Datagram Protocol

- No overhead but also not as reliable
- Doesn't check for errors
- Good for real-time applications like live-broadcast or video or online video games
- **Multiplexing/Demultiplexing**
 - does **not** require info about the size/control of the connection, since its connectionless **[dest IP, dest Port]**

Network Layer

- Responsible for packet forwarding
- Routing through intermediate routers

Link Layer (briefly)

- Error-checking
- Flow control

- Access control

Congestion Control

- Controls traffic entry into a telecommunications network to avoid congestive collapse
- Avoids oversubscription of any of the processing/link capabilities

Different Types of Delays

- **Processing delay:** time it takes router to examine the packet header and determine where to direct the packet
- **Queuing Delay:** time a packet spends in the output buffer waiting to be transmitted.
 - based on amount of congestion at the time.
 - **traffic intensity:** $(L \cdot a)/R$
 - $L = \# \text{ bits}$
 - $R = \text{rate}$
 - $a = \# \text{ of packets}$
- **Transmission Delay:** time it takes to push a packet's bits into the link
 - $L \text{ bits} / R(\text{rate})$
 - based on the size of the packet
- **Propagation delay:** time it takes for bits to physically travel from point A to B
 - based on the medium/distance between points

Concepts of Transport Layer (no computations)

1 rigorous computational problem

What are all the different formulas we've learned to date?

- **Propagation time/delay**
 - $\text{Distance} / \text{Prop_Rate}$
- **Header Transmission Time**
 - $\text{Size} / \text{Link-Speed}$
- **Data Transmission Time**
 - $\text{Data-packet-size} / \text{Link-Speed}$

- **Transmission Time/Delay**

- Number of bits/ Rate of transmission

- **Distribution Times**

- **Client/Server** = $\text{Max}(NF/Us, F/d_{\min})$

- $N = \# \text{ of peers}$
- $F = \text{length in bits}$
- $Us = \text{Upload speed of server}$
- $d_{\min} = \text{lowest download rate of any of the peers}$

- **P2P** = $\text{Max}(F/Us, F/d_{\min}, NF/Us + \text{sum}(U_i))$

- $\text{sum}(U_i) = \text{upload speed for all peers}$

- **Bandwidth-delay:** max number of bits the link has at any time

- $R * d_{\text{prop}}$

- $d_{\text{prop}} = \text{round trip propagation delay} = \text{distance} * \text{prop_rate}$

- **Average Throughput:** if file has F bits, transfers over T seconds:

- $\text{AvgThroughput} = F/T$

- **End-to-End delay:** $d_{\text{E2E}} = (\# \text{ routers}) * (d_{\text{processing}} + d_{\text{transmission}} + d_{\text{propagation}})$

- **Nodal Delay:** $d_{\text{nodal}} = d_{\text{processing}} + d_{\text{transmission}} + d_{\text{propagation}}$

- any others???

Conversions:

- **Bytes -> Bits:** (Bytes * 8): 1 byte = 8 bits

- **Kb** = 1,000 bits

- **Mb** = 1,000,000 bits

- **Gb** = 1,000,000,000 bits

Circuit-Switching:

Allocates a set amount of memory per operation.

- Phone call would be circuit-switching because we are transmitting and receiving a steady amount of data.

- It's not at random, the connection is a full circuit. This is much more expensive but also very reliable.

- We are moving away from circuit switching because it is too expensive
- Switches are reserved

Packet-Switching

- Message gets broken into smaller data packets
- Gets sent through the most efficient route as circuits become available
- Less reliable than Circuit Switching

Persistent vs Non-persistent

- In a persistent connection, the server always keeps its connection open (unless there is a timeout IE the internet is down)
 - less reliable
- In a non-persistent connection, the client establishes a connection with the server, request data from a server, the server responds to client, the connection is closed.
 - less efficient

Winter 2015 Midterm Questions

T/F

1. In the Internet Protocol (IP) stack, the network layer can guarantee the reliable transmission from the source to the destination.
 1. False. IP uses UDP to transfer data
2. Virtual circuits (ie circuit switching) always has longer transmission delays than packet switching
3. TCP segment headers are always 20 bytes in length
 1. False. Can be greater than 20 (if "options" section is used)
4. In a packet-switched network, all routers must keep information for all end-to-end connections
 1. False. routers don't store any client information.
5. The DNS uses UDP transport to exchange messages
 1. True
6. SMTP uses UDP transport to exchange messages
 1. False
7. The Selective Repeat (SR) does not require buffers on both the sender and receiver
8. TCP congestion control is used to prevent the sender from overflowing the receiver

FILL IN THE BLANK

1. Link Layer is responsible for end-to-end flow control
2. Link layer is responsible for reliable transfer of information
3. Transport layer is responsible for creating data segments
4. Network Layer is responsible for routing decisions
5. Physical Layer is responsible for transmission of raw bit stream
6. Transport layer is responsible for defining sockets and transport mechanisms: **this should be applicatoin**

SHORT ANSWER

1. Briefly describe the difference between a "push" and "pull" protocol. Give an example for each.
 1. Push protocol: instantly streams data to you, like a cell-phone call.
 2. Pull protocol: send/receive. You ask for a webpage, you get one in response.
2. Briefly define each of the seven layers in the ISO/OSI layering architecture. Describe how the IP protocol is derived from this.
 - 1.
3. Define the connections and port numbers that FTP uses. What is the purpose of each of them? Which connections are persistent and non-persistent?
4. What information is necessary for UDP mutiplexing/ demultiplexing? What information is necessary for TCP multiplexing/demultia

Spring Midterm

Choose only one answer.

You will not receive any points if you choose multiple answers.

1. In the OSI networking stack, routing is performed by the
 - a) Session Layer
 - b) Network Layer**
 - c) Transport Layer
 - d) Data-Link Layer
 - e) None of these choices
2. Four bits are used for sequence numbering in a sliding window protocol used in a computer network. What is the maximum window size? a) 4 b) 8 c) 15 **d) 16** e) None of these choices
3. Which of the following TCP/IP protocols is used for remote terminal connection services?
 - a) TELNET**
 - b) FTP
 - c) RARP
 - d) UDP
 - e) None of these choices
4. Which protocol is used for sending email on the Internet?
 - a) SMTP**
 - b) SMPP
 - c) SNMP
 - d) FTP
 - e) None of these choices

5. Which one of the following uses the greatest number of layers in the TCP/IP stack?
a) Switch b) Repeater c) Router **d) End Host** e) None of these choices
6. What is the default port number for HTTP? a) 21 **b) 80** c) 25 d) 8080 e) None of these choices
7. A basic telephone network is an example of a) Packet Switching b) Cell Switching **c) Circuit Switching** d) Message Switching e) None of these choices
8. Which one of the following is used to communicate between different networks? a) ADSL b) HDSL **c) Gateway/Router** d) Modem e) None of these choices
9. Which of the following OSI layers corresponds to the layer where TCP operates in the TCP/IP stack? a) Network Layer b) Data-Link Layer c) Session Layer **d) Transport Layer** e) None of these Choices
10. TCP uses the following mechanism for increasing the congestion window size (cwnd): a) additive increase additive decrease **b) additive increase multiplicative decrease** c) multiplicative increase additive decrease d) multiplicative increase multiplicative decrease e) None of these choices

Problem #2: (6 points, 1 point each) Determine whether each of the following statements is True (T) or False (F). No explanation is necessary; partial credit will not be awarded.

1. All nodes connected to the Internet must implement UDP. (F)
2. Media access control is a function of the data-link layer. (T)
3. Forward Error Correction (FEC) can be more efficient than Automatic Repeat Request (ARQ) in a broadcast environment with many receivers. (T)
4. End hosts implement transport layer protocols but routers do not. (T)
5. In a packet switched network, all packets belonging to the same transport layer session must follow the same route. (F)
6. Peer-to-Peer (P2P) systems are scalable because available resources increase with a quadratic factor with each new member that joins the P2P system. (F)

Problem #3: (6 points) Answer the following questions regarding the Application Layer. (1 points)

1. Briefly describe what HTTP is and sketch its operation using a simple figure (i.e., the typical messages exchanged during operation of HTTP).

The HyperText Transfer Protocol (HTTP) is a stateless application layer protocol. HTTP is used to transfer web content between a browser application (client) and an HTTP server. All web content is identified by a URL. HTTP is a request response protocol that uses TCP for assured delivery. HTTP uses ASCII encoded headers. The HTTP GET command retrieves HTML files and other objects. The GET header includes the URL of the object and other optional fields such as capability, language, and so on. The response includes a response header with a response code (code 200 is OK and 404 is page not found). Other commands include POST and HEAD.

```
+-----+
+ +-----+ | Client |----- Request ----->| Server ||<----- Response -----| +-----+ +-----+
(1/2 point)
```

2. We say that FTP has out-of-band control. What do we mean by that? We mean that commands and data flow across different TCP connections. (1 point)
3. What is the minimum value for the timeout of a reliable transmission protocol? Why is there a minimum value? One RTT; anything shorter would timeout before the ACK had a chance to arrive. (1/2 point)
4. What is the difference between congestion control and flow control? Congestion control prevents overrunning buffers in the network, while flow control prevents overflowing the end hosts (or receivers). (3 points)
5. Consider 5 users (call them A, B, C, D, and E) that have connections on a single 10 Mb/s link. Connections last for several minutes or perhaps even hours (i.e., they are bounded in time). Assume that user A requests 2 Mb/s for its connection, B requests 1 Mb/s, C requests 3 Mb/s, D requests 4 Mb/s, and E request 10 Mb/s. Note that the total requested bandwidth (20 Mb/s) exceeds the link bandwidth (10 Mb/s). Describe at least three ways of fairly allocating the 10 Mb/s to the five users. Carefully discuss/describe what is "fair". Carefully discuss the tradeoffs between your different definitions of "fair". Multiple definitions of fair are possible. One definition of fair can be to allocate resource proportional to need. So, A gets $(2/20)(10)$ Mb/s = 1.0 Mb/s B gets $(1/20)(10) = 0.5$ C gets $(3/20)(10) = 1.5$ D gets $(4/20)(10) = 2.0$ E get $(10/20)(10) = 5.0$ This definition of fairness rewards over-requesting (cheating) to get a larger share against other users. We note that in this example, no user is 100% happy.

http://www.ics.uci.edu/~keldefra/teaching/spring2013/uci_cs132/exams/cs132_midterm_sol_ked.pdf

CS 132 Outline

Overview

- Terminology

- Network Edge
- Internet Service Provider (ISP)
- Network Core

- Latencies / Rates

- Transmission Rate
- Processing Delay
- Queueing Delay
- Transmission Delay
- Propagation Delay

- Packet Switching vs Circuit Switching

- Time Division Multiplexing
- Frequency Division Multiplexing

- Pros / Cons for each
 - Â€ OSI 7-Â€layer stack
 - Â€ IP Stack

Application Layer

- Â€ Architectures
 - Client-Â€server
 - Peer-Â€to-Â€peer (P2P)
 - Â€ Ports
 - Â€ Processes
 - Â€ Sockets
 - Â€ Common transport services applications can use
- TCP / UDP
 - Â€ Applications
- HTTP
 - â–a Cookies
 - â–a Web Caching (Proxy Server)
- FTP
 - â–a Control Connection
 - â–a Data Connection
- SMTP
 - â–a Push vs pull protocol
- DNS
 - â–a Hierarchy
- Root DNS
- Top-Â€level Domain (TLD)
- Authoritative
- Local DNS

Recursive vs Iterative Queries

- P2P applications
 - Distributed Hash Tables
 - P2P vs. Client / Server Distribution Time
- Overlay Networks

Transport Layer

- Multiplexing / Demultiplexing
- TCP vs UDP
 - UDP
- Characteristics
 - Memory required
 - Lack of congestion control
- Segment Structure
 - Source / Destination Port
 - Length in bytes (size of data + header)
 - Checksum
 - Reliable Transport
- Stop-and-wait protocol
- Pipelining
- Go-Back-N
- Selective Repeat
- Piggybacking
- Differences between each of these
 - TCP
- Characteristics
 - Memory required
 - Congestion Control vs Flow Control
- Segment Structure

• Source / Destination Port

• Checksum

• Sequence / ACK number

• Receive Window Size

• Header length

• Options

• Flags

- ACK, RST, SYN, FIN, PSH, URG

- Timeout Estimation

- TCP Optimizations

 - Doubling Timeout Interval

 - Fast Retransmit

 - Flow Control

- Receiver Variables

 - LastByteRead, LastByteRcvd, RcvBuffer, rwnd

$$rwnd = RcvBuffer - [LastByteRcvd - LastByteRead]$$

-

- Sender Variables

 - LastByteSent, LastByteAcked

 - $LastByteSent - LastByteAcked \leq rwnd$

 - Congestion Control

- Congestion Window (cwnd)

- Slow Start

- Congestion Avoidance

- Fast Recovery

- TCP Tahoe vs TCP Reno

- TCP Connection Management

 - Three-way handshake

 - TCP Connection Teardown

Network Layer

-♦ Forwarding vs Routing

Forwarding - single router, moves data from an input, port to an output

Routing - Determines path taken from sender to receiver

-♦ Packet vs Circuit Switching

Packet/Datagram - Resources are first come, first serve (no state to keep track of)

Circuit Switched/Virtual Circuit - Resources are reserved from source to destination

-♦ Router

Each router has a forwarding table that maps destination addresses to link interfaces; when a packet arrives at the router, the router uses the packet's destination address to look up the appropriate output link interface in the forwarding table. The router then intentionally forwards the packet to that output link interface. All destination address 32 bits

Input ports - Terminate incoming physical link on physical layer. Look up function with forwarding table. Decides if arriving packet will be forwarded with switching fabric.

Switching fabric - Connects router input ports to output ports. Encapsulated within router (network inside network router)

Output ports - Stores packets received from switching fabric and transmit on outgoing link on physical and link layer.

Routing processor - Executes routing protocols and maintains routing tables with link state information and computes forwarding table.

• Router Forwarding Tables

Packet/Datagram - No connection state information, but maintains forwarding state information in their forwarding tables. Time depends on routing algorithm to update table about every 1-5 minutes

Circuit Switched/VC network - a forwarding table in a router is modified whenever a new connection is set up through the router or whenever an existing connection through the router is torn down

• Longest Prefix Matching

Finds the longest matching entry in the table and forwards the packet to the link interface associated with the longest prefix match.

• Packet Scheduling (Active Queue Management)

Packet loss occurs when queue gets too big and router runs out of memory.

Packet scheduler at the output port must choose one packet among those queued for transmission. This selection might be done on a simple basis, such as first-come-first-served (FCFS) scheduling, or a more sophisticated scheduling discipline such as weighted fair queuing (WFQ), which

• Weighted Fair Queueing (WFQ)

Shares the outgoing link fairly among the different end-to-end connections that have packets queued for transmission.

• Drop Tail

Decides to drop arriving packet or remove queued packet for room.

• Provides some level of Quality of Service (QoS)

• Internet Protocol (IP) Datagram Header

20 bytes of header with no options. 40 bytes if carries TCP(nonfragmented)

- Version Number - 4 bits, Interprets remainder of datagram by version (ipv4, ipv6)
- Header Length- 4 bits determines where data begins, usually 20 byte header
- Type of Service (ToS)- Real-time(IP) or non-real time traffic(FTP)
- Datagram Length- 16 bits; Total length of IP datagram(header+data) Max 65535 bytes, usually less than 1500 bytes.
- Identifier, Flags, Fragmentation Offset - ipv6 does not allow fragmentation
- Time-to-Live (TTL)- Ensure datagrams don't circulate forever. Decrements by one each time until 0 and drops datagram.
- Protocol - Indicates transport-layer protocol to which datagram should be passed. Used when IP datagram reaches destination. (ie 6 = TCP, 17 = UDP)
- Checksum - Header checksum detects bit errors in received datagram. Computed by 2 bytes in header as a number and summing numbers using 1s complement. Router discards datagram if does not match.

IP header is checksummed at the IP layer, while the TCP/UDP checksum is computed over the entire TCP/UDP segment.

TCP/UDP and IP do not necessarily both have to belong to the same protocol stack. TCP can, in principle, run over a different protocol (for example, ATM) and IP can carry data that will not be passed to TCP/UDP

- Source / Destination IP addresses
- Options (variable, rarely used) - Allows IP header to be extended, not in ipv6
- Data (payload)- Contains TCP/UDP to be delivered, can carry other data like ICMP

-♦ IP Data Fragmentation

- MTU – largest link-layer frame that can be sent on the link

Maximum amount of data link-layer frame can carry. Useful when forwarding table determines outgoing link has MTU smaller than IP datagram length. Fragments datagram into smaller IP datagrams, encapsulated in separate link-layer frame and send frames over outgoing link. Smaller datagrams are fragments.

- Fragments packets based on the MTU along the path

ipv4 has identification flag and fragmentation offset in IP header.

Identification number increments for each datagram sent. Used to determine if fragments of same datagram.

Last fragment flag - set to 0, 1 for all other fragments.

Offset field - specifies where fragment fits to check for missing fragments

-♦ IPv4 Addressing

Host typically has one single link on network to send datagram on IP.

Interface - boundary between host and physical link. (router and link)

- Applies to host interfaces

IP requires host and router interface to have own IP address.

IP address associated with interface, not host or router with interface

Network connecting 3 host interfaces and one router forms subnet/network.

- Dotted Decimal Notation

32 bits long, 4 bytes. 2^{32} possible addresses

Each interface on host and router must have globally unique IP address

determined by subnet it is connected to.

– 0.0.0.0 – 255.255.255.255

- Classless Interdomain Routing (CIDR)

Generalizes notion of subnet addressing. 32 bit IP address divided into two and again.

a.b.c.d/x; x = prefix (number of bits in first part of address)

– Network prefix vs host IP address

when a router outside forwards a datagram to address inside, only the leading x bits of the address need be considered. Reduces the size of the forwarding table in these

routers, since a single entry of the form a.b.c.d/x will be sufficient to forward packets to any destination within the organization.

- Classful Addressing

Network portions of an IP address were constrained to be 8, 16, or 24 bits in length, an addressing scheme known as classful addressing, since subnets with 8-, 16-, and 24-bit subnet addresses were known as

class A, B, and C networks, respectively.

• Class A: /8

• Class B: /16

supports up to 65,534 hosts

• Class C: /24

Supports up to $2^{24} - 2 = 16,777,214$ hosts (two of the $2^{24} = 16,777,216$ addresses are reserved for special use) too small for many organizations.

- Broadcast Address

ICANN allocates IP addresses and manages DNS root servers.

• 255.255.255.255

• Dynamic Host Configuration Protocol (DHCP)

Client-server protocol and plug&play capability. DHCP allows a host to obtain (be allocated) an IP address automatically instead of assigning individual IP addresses from IP block. DHCP also allows a host to learn additional information, such as its subnet mask, the address of its first-hop router (often called the default gateway), and the address of its local DNS server.

4 step process protocol:

- DHCP Server Discovery- done using a DHCP discover message, which a client sends within a UDP packet to port 67. The UDP packet is encapsulated in an IP datagram.
- DHCP Server Offer- DHCP offer message that is broadcast to all nodes on the subnet, again using the IP broadcast address of 255.255.255.255. Each server offer message contains the transaction ID of the received discover message, the proposed IP address for the client, the network mask, and an IP address lease time—the amount of time for which the IP address will be valid. (ranges from hours or days)
- DHCP Request- choose from among one or more server offers and respond to its selected offer with a DHCP request message, echoing back the configuration parameters.
- DHCP ACK-Server responds to the DHCP request message with a DHCP ACK message, confirming the requested parameters.

Once the client receives the DHCP ACK, the interaction is complete and the

client can use the DHCP-allocated IP address for the lease duration.

-Â€ Network Address Translation (NAT)

- Mapping of "private" IP and port numbers to "public" IP and port numbers

-Â€ Internet Control Message Protocol (ICMP)

- "Types" and "Codes"
- Common examples

" Ping

" Traceroute

-Â€ IPv4 vs IPv6

- Address space
- Constant header length
- Removal of Fragmentation
- Removal of Checksum
- Problems with migrating from IPv4 to IPv6

-Â€ Routing Algorithms

- Global Routing Algorithms

" Link- state algorithms

" Dijkstra's Link State algorithm

- Decentralized Routing Algorithms

" Distance- vector algorithm (Bellman- Ford Equation)

" Count- to- Infinity Problem

- Poison Reverse
- Hierarchical Routing

" Autonomous Systems

" Gateway Routers

" Hot Potato Routing

" Intra- AS Routing

- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)

• Inter-AS Routing

- Border Gateway Protocol (BGP)
- External BGP (eBGP) sessions
- Internal BGP (iBGP) sessions

Link Layer

- Responsible for sending data across individual links

- Node: Computational device running a link-layer protocol (hosts, routers, switches)

- Link: Communication channels connecting adjacent nodes.

- Network Adapter / Network Interface Card (NIC)

- Where hardware meets software

- Error Detection and Correction

- Parity Bits

• Even / Odd Parity

• Two-Dimensional Parity Checks

- Forward Error Correction (FEC)

- Checksumming

- Cyclic Redundancy Check (CRC)

- Multiple Access Links and Protocols

- Channel Partitioning

• Time Division Multiplexing

• Frequency Division Multiplexing

- Random Access Protocols

- Slotted ALOHA

- ALOHA

- Carrier Sense Multiple Access / Collision Detection (CSMA / CD)

- Binary Exponential Backoff

- Taking Turns Protocol

- Polling Protocol

- Switched Local Area Networks

- 48 bits

$$\hat{a}^{-a}\hat{a}^{-a}00\hat{\in}00\hat{\in}00\hat{\in}00\hat{\in}00\hat{\in}00\hat{\dagger}\hat{\dagger}\text{FF}\hat{\in}\text{FF}\hat{\in}\text{FF}\hat{\in}\text{FF}\hat{\in}\text{FF}\hat{\in}\text{FF}\hat{\in}\text{FF}$$

- $$\hat{a}^{-a}\hat{a}^{-a}FF-\hat{A}\hat{A}\in FF-\hat{A}\hat{A}\in FF-\hat{A}\hat{A}\in FF-\hat{A}\hat{A}\in FF-\hat{A}\hat{A}\in FF$$

- ## â–aâ–a ARP Table

Forwarding packet within a subnet

Forwarding packet to an external subnet

Link Layer Switches

- Switch Filtering vs Broadcasting

Switches vs Routers

- Pros / Cons for each.